Design and Implementation of an Auto-Ranging 12-Bit Impedance Analyzer and Logic Tester

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1.Introduction

Apex v1.0 is an advanced, dual-microcontroller diagnostic instrument designed for highprecision, auto-ranging impedance measurement and analog signal analysis. Built around an ESP32 master controller and an ATmega328P co-processor, the system synthesizes DC-biased sinusoidal test signals using an AD9833 waveform generator to stimulate a Device Under Test (DUT). To achieve a wide dynamic measurement range without sacrificing resolution, the architecture employs custom-built Programmable Gain Amplifier (PGA) and Transimpedance Amplifier (TIA) stages dynamically scaled by a CD4052 analog multiplexer network. By capturing peak-to-peak signal attenuation through an MCP3208 12-bit ADC and processing the differential telemetry in real time, this project aims to deliver a robust, fully integrated PCB platform capable of autonomous component evaluation, error correction, and precise resistance calculation at the edge.

2.System Architecture

2.1Block Diagram

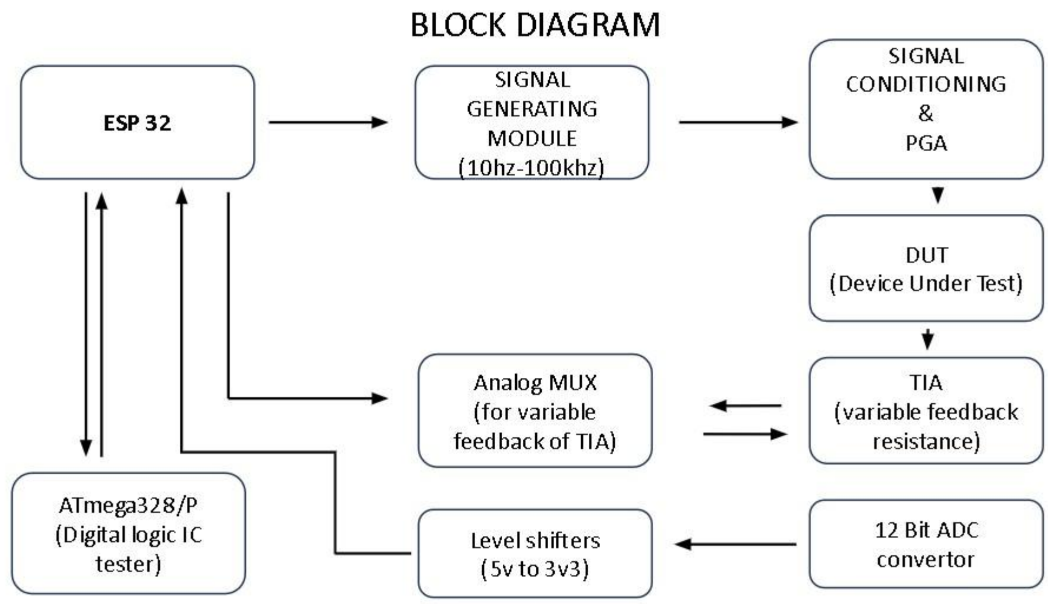


Fig 1

2.2 Component Selection & Rationale

2.1 Analog Front-End (AFE): MCP6022 vs. Standard Op-Amps

The signal conditioning, Programmable Gain Amplifier (PGA), and Transimpedance Amplifier (TIA) stages dictate the baseline accuracy of the entire instrument. The MCP6022 was specifically selected over general-purpose alternatives (such as the MCP6004) due to its superior slew rate and significantly lower input offset voltage. In an architecture dealing with precise 2.2V–2.8V synthesized waveforms and micro-volt variations across a Device Under Test (DUT), compromising on slew rate introduces severe phase distortion, while high input offset directly corrupts the impedance transfer function.

2.2 Data Acquisition: MCP3208 vs. ESP32 Internal ADC

While the ESP32 features an internal ADC, its silicon is notoriously non-linear, exhibits high internal noise, and suffers from poor channel-to-channel isolation, making it entirely unsuitable for high-fidelity instrumentation. To guarantee measurement integrity, the MCP3208 was integrated. This provides a dedicated, true 12-bit resolution data pipeline operating at 100 ksps over an isolated SPI bus. Utilizing this external ADC alongside a stable external voltage reference ensures the absolute linearity required for the auto-ranging measurement algorithms.

2.3 Logic Level Translation: 74HCT125 & Resistive Scaling

The dual-voltage architecture of Apex v1.0 creates a strict boundary between the 3.3V digital logic of the ESP32 and the 5V analog peripherals required to maximize the ADC's dynamic range. To protect the silicon and maintain high-speed SPI signal integrity, a 74HCT125 quad buffer is deployed to actively translate the 3.3V MOSI, CS, and SCK signals from the ESP32 up to the 5V domain. Conversely, precision resistive voltage dividers safely step down the 5V MISO return signals to the 3.3V ESP32 inputs, preventing overvoltage stress. Additionally, digital control signaling for the 5V CD4052 analog multiplexer is handled via discrete BJT-based inverter circuits. This configuration effectively level-shifts the 3.3V ESP32 GPIOs to 5V while inherently inverting the control logic, a parameter directly accounted for in the firmware execution matrix.

2.4 Digital Logic Tester: ATmega328P & ZIF Routing Matrix

To expand Apex v1.0 into a comprehensive diagnostic instrument, a dedicated digital logic IC tester was integrated utilizing the ATmega328P as a secondary co-processor. Offloading the logic validation to the 5V-native ATmega328P ensures deterministic execution of bit-banging test vectors without interrupting the ESP32 master’s high-speed SPI data acquisition or RTOS scheduling. The physical interface employs a 16-pin ZIF socket coupled with a hardware routing matrix constructed from discrete 1P3T (Single Pole, Triple Throw) switches for each pin. This mechanical topology allows dynamic configuration of the power rails to accommodate varying logic IC pinouts (e.g., 74-series or CD4000-series logic). Setting a switch to the left state ties the respective ZIF pin to VCC, the right state ties it to GND, and the center state routes it directly to the ATmega328P GPIOs for logic stimulus and reading. Upon completing the truthtable verification, the ATmega328P transmits the pass/fail diagnostic data back to the ESP32 master via an asynchronous serial link (UART) for final processing and OLED display.

3 Hardware Prototyping & Physical Limitations

3.1 Breadboard Iteration: SPI Signal Degradation

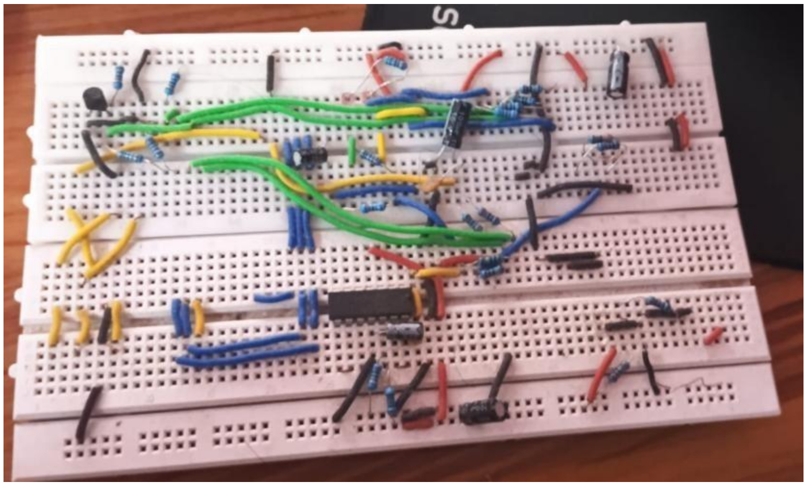
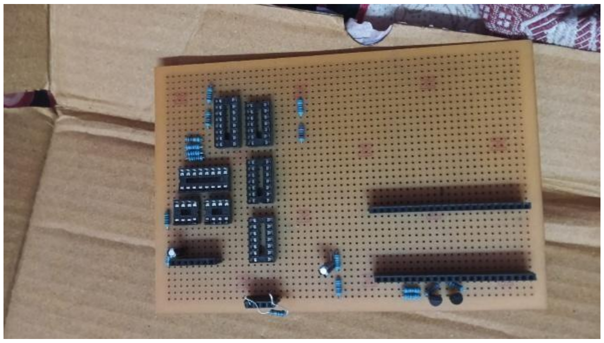


Fig 2(all chips have been removed for next iteration stage)

Initial validation of the analog front-end and logic translation was conducted on a breadboard. However, the parasitic capacitance and stray inductance inherent to breadboard tracks severely corrupted the 100 ksps SPI bus between the ESP32 and the MCP3208. The ringing on the clock (SCK) and data lines, combined with the mechanical instability of spring contacts, resulted in erratic data acquisition and excessive ground bounce, necessitating a soldered prototype.

3.2 Perfboard Iteration: Routing Density & Micro-Shorts



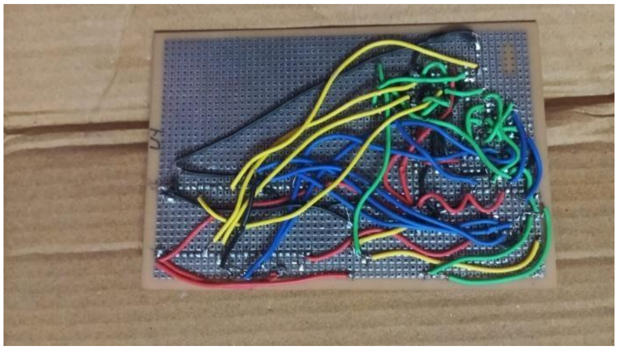


Fig 3

To resolve the mechanical instability of the breadboard, a complete point-topoint soldered perfboard prototype was constructed. While this eliminated loose connections, the sheer volume of wiring required to bridge the ESP32, ATmega328P, AD9833, multiplexers, and high-speed ADCs created an unsustainable routing density. The tightly packed wire matrix ultimately resulted in microscopic short circuits (solder bridges) between adjacent logic pins.

Due to the layered nature of point-to-point wiring, isolating and repairing these micro-shorts proved practically impossible without destructively dismantling the prototype.

3.3 The 4-Layer PCB Rationale: Signal Integrity & Manufacturability

The catastrophic failures of both the breadboard (signal integrity) and perfboard (routing complexity and micro-shorts) dictated an immediate transition to a custom Printed Circuit Board (PCB). A 4-layer stackup was explicitly chosen over a standard 2-layer design to achieve two critical objectives:

Routing Simplification:

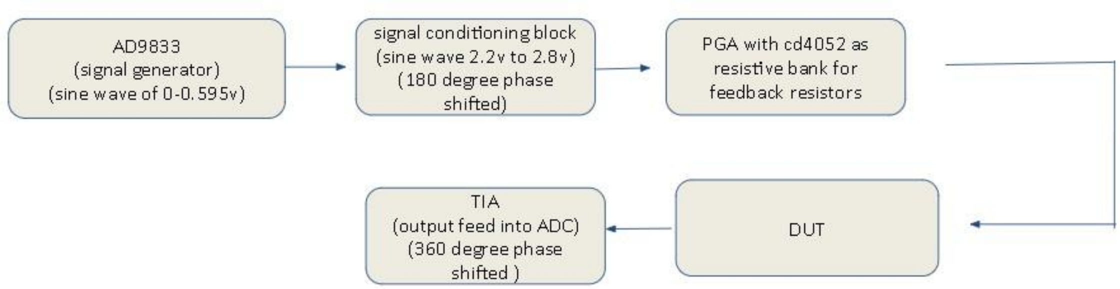
By dedicating internal layers to uninterrupted Ground (GND) and Power (3.3V/5V) planes, power delivery routing was eliminated from the surface layers, drastically reducing trace density and completely eliminating manual wiring faults.

Signal Integrity:

The continuous internal ground plane provided a direct return path for high-speed digital signals (SPI bus), minimizing electromagnetic interference (EMI) and crosstalk with the sensitive 2.2V–2.8V analog front-end.

4 PCB Design & Manufacturing

4.1 Schematic Capture & EDA Workflow



Programmable Gain Amplifier (PGA) Schematic & Signal Flow

Signal Injection and Phase Dynamics: Following the initial signal conditioning block, the synthesized waveform enters the PGA stage as a DC-biased sine wave swinging between 2.2V and 2.8V, bearing a 180-degree phase shift relative to the AD9833 origin. The primary objective of the PGA is to dynamically scale this excitation signal to maximize the measurement resolution across the Device Under Test (DUT) without saturating the subsequent Transimpedance Amplifier (TIA) stage.

Dynamic Gain Control via CD4052: The core of the PGA is built around an MCP6022 precision operational amplifier. Rather than utilizing a fixed feedback loop, the amplifier's gain is actively modulated using a CD4052 analog multiplexer configured as a switchable resistive bank. The multiplexer routes the feedback path through one of four discrete resistor channels. These channels are selected via the A_PGA and B_PGA control lines, which are driven by the ESP32 through BJT-based logic inverters to safely bridge the 3.3V to 5V domain boundary.

Calibration and Gain Coefficients: To ensure absolute mathematical precision in the firmware's auto-ranging transfer function, the nominal feedback resistors were physically characterized. The precise measured values are hardcoded into the system for algorithmic scaling:

Channel 0 (Y_0): 1.496 kΩ (Nominal 1.5 kΩ)

Channel 1 (Y_1): 2.980 kΩ (Nominal 3.0 kΩ)

Channel 2 (Y_2): 4.660 kΩ (Nominal 4.5 kΩ)

Channel 3 (Y_3): 6.700 kΩ (Nominal 6.0 kΩ)

This discrete, four-stage amplification allows the instrument to step through gain profiles autonomously. The amplified output from this stage is simultaneously tapped and fed into Channel 3 of the external MCP3208 ADC to record the baseline excitation voltage, while the primary current path continues forward to drive the DUT.

4.2 KiCad Schematic

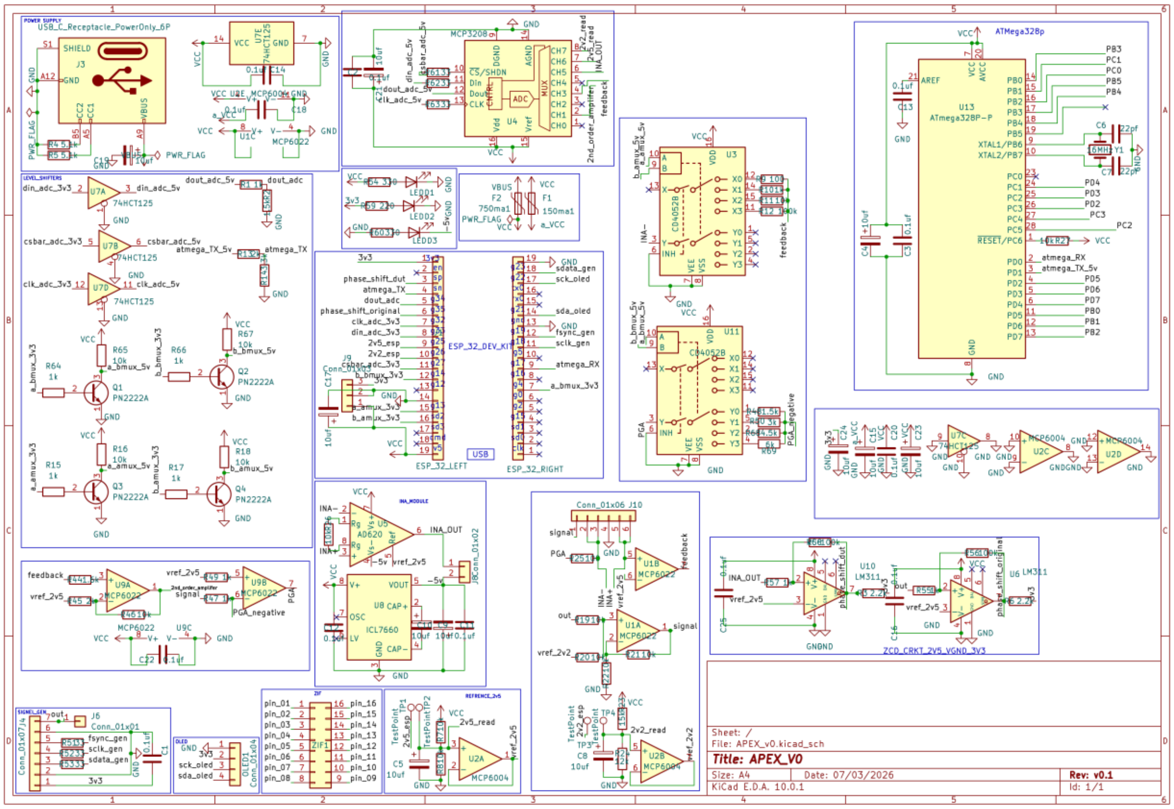


Fig 5

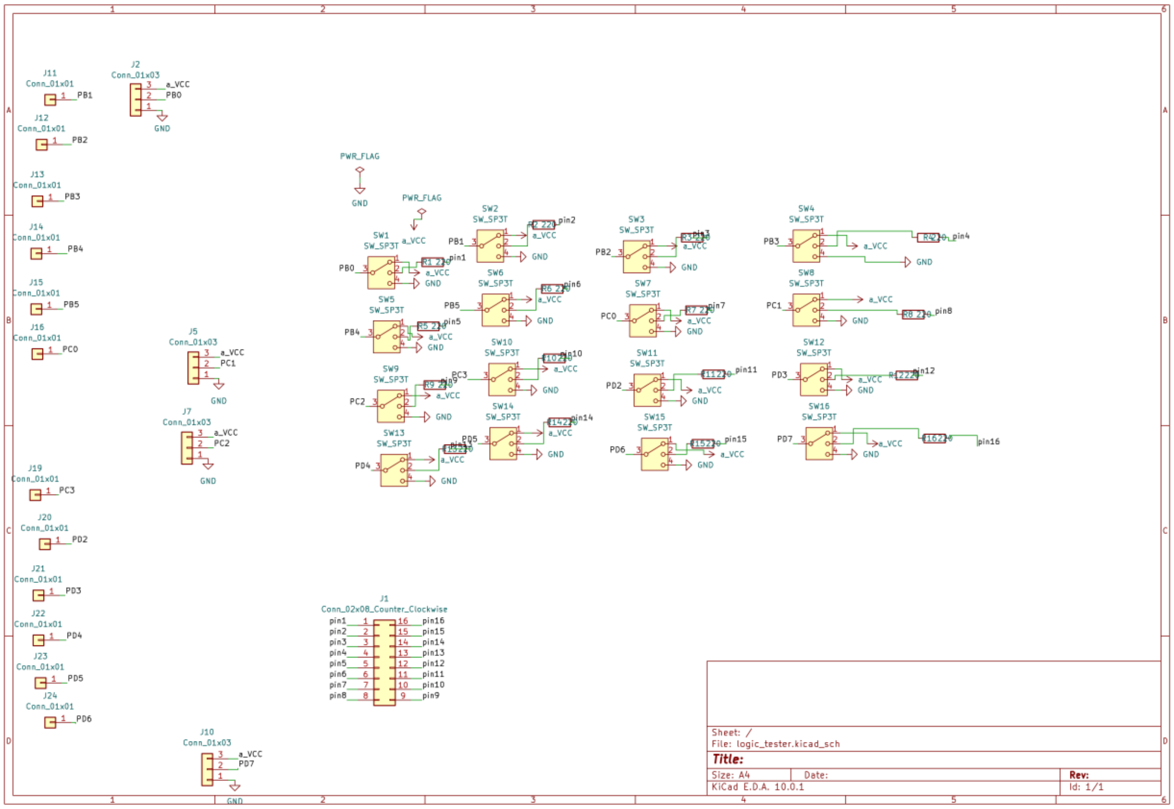
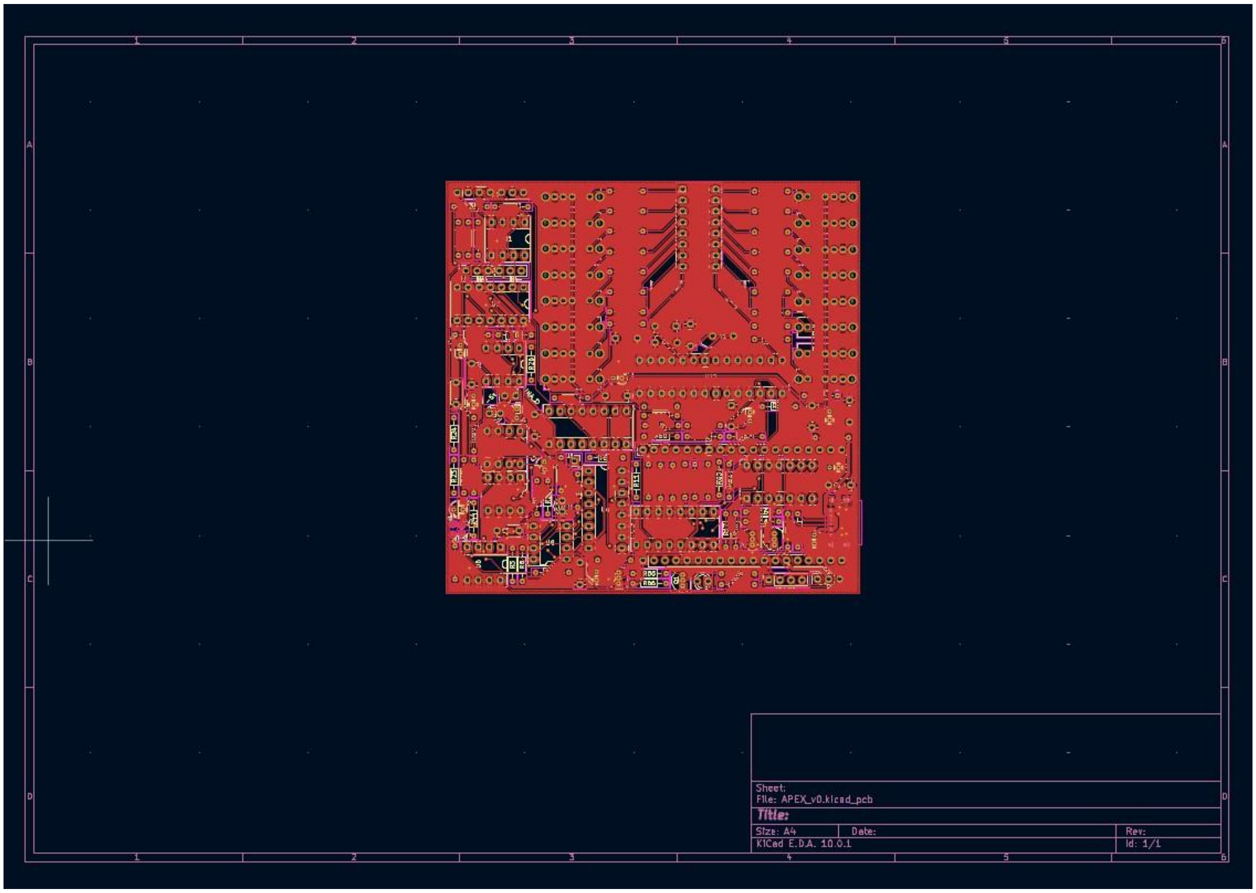


Fig 6

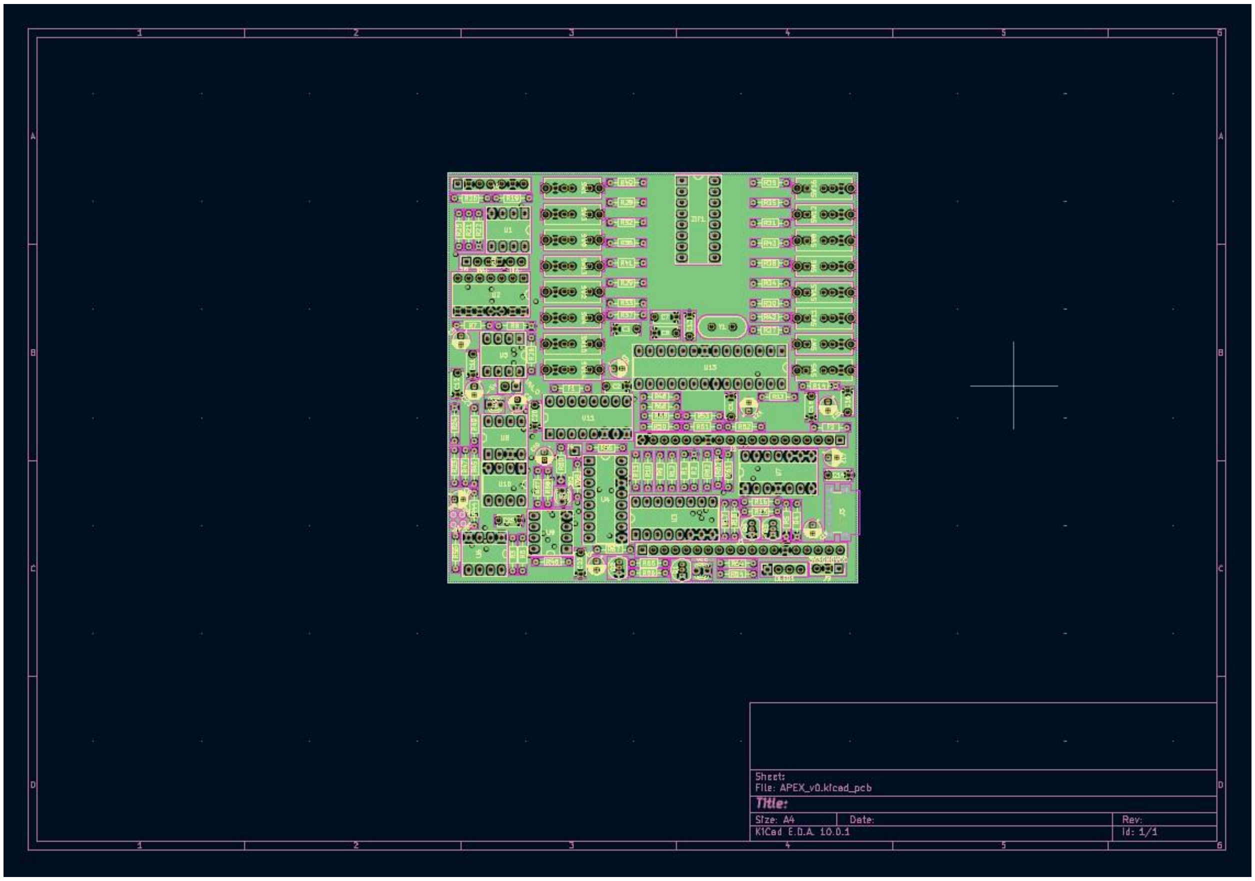
Schematic Capture & System Overview

The electronic design and schematic capture for Apex v1.0 were executed using KiCad EDA. To manage the complexity of the 15-in-1 architecture, the system was designed using a modular, block-level approach. The primary schematic isolates the core functional domains: the ESP32 master controller, the ATmega328P co-processor, the power delivery/level-shifting network, and the highly sensitive analog front-end (AFE) containing the MCP3208 ADC and CD4052 multiplexers. By logically separating the 3.3V digital logic from the 5V analog measurement paths at the schematic level, signal interference is minimized. A dedicated secondary schematic manages the digital logic IC tester, mapping a 16-pin ZIF socket through a hardware routing matrix of discrete 1P3T switches. This matrix dynamically routes VCC, GND, or ATmega328P GPIO stimulus to any pin on the DUT, ensuring the schematic remains readable and structurally organized for the physical PCB layout phase.

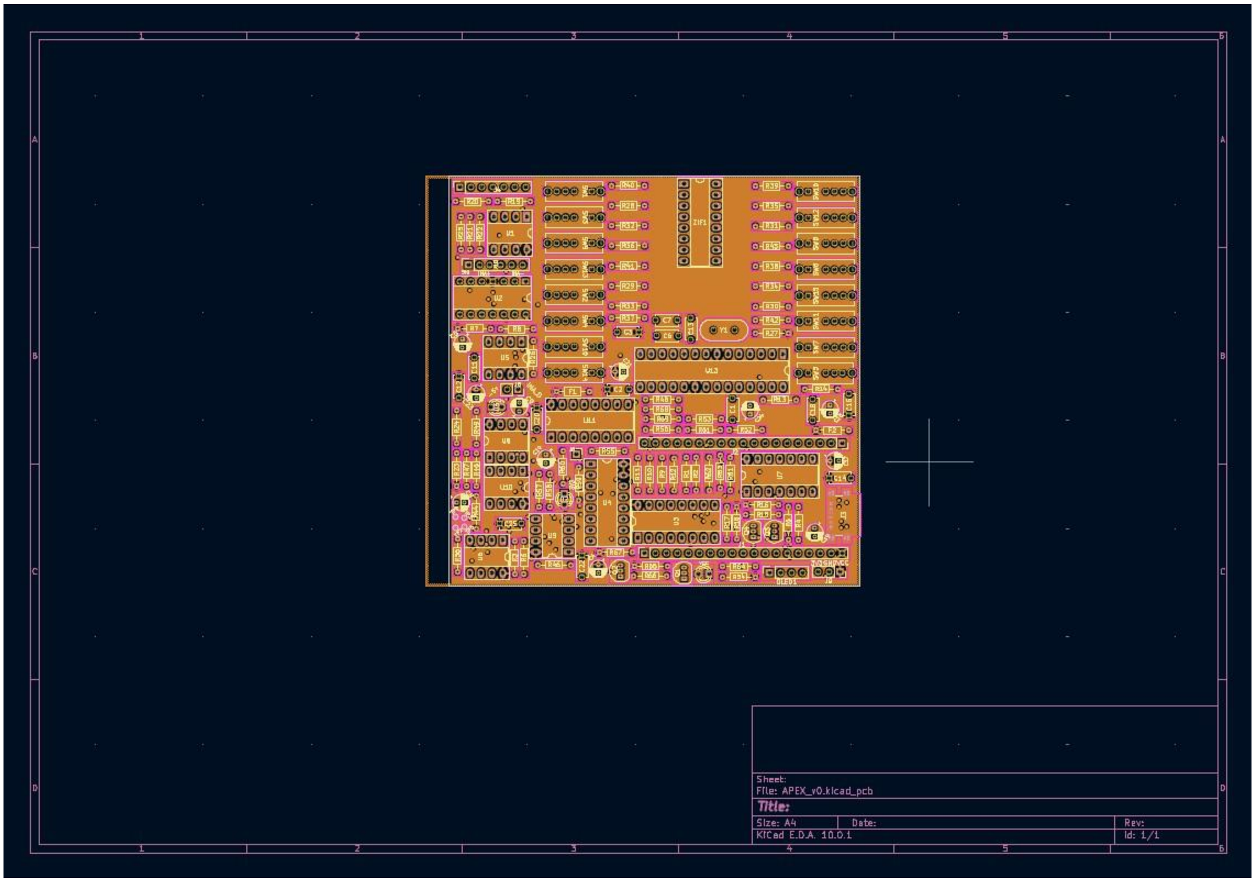
4.3 KiCad PCB Layers



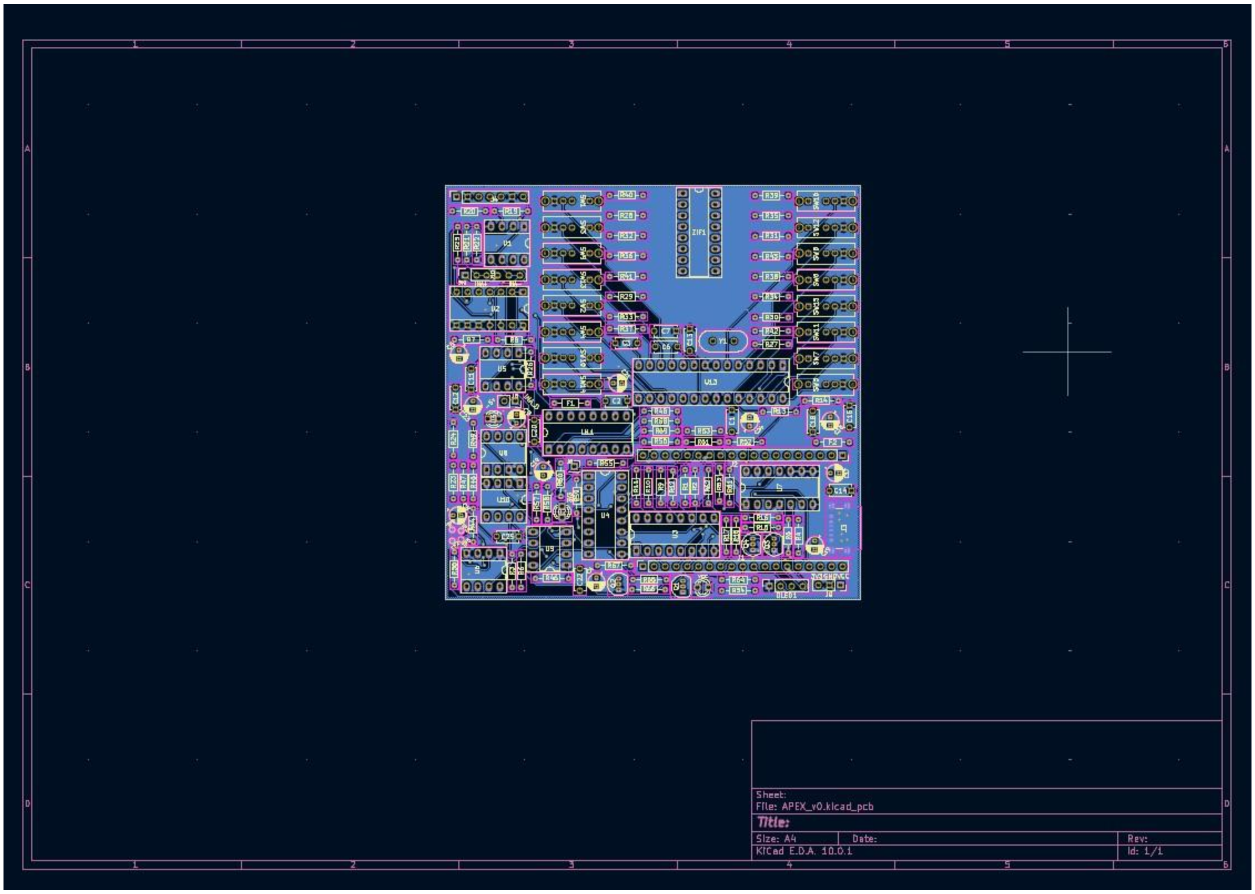
Layer 1: primary signal layer



Layer 2:Stable Ground layer to avoid ground loop and EMI noise

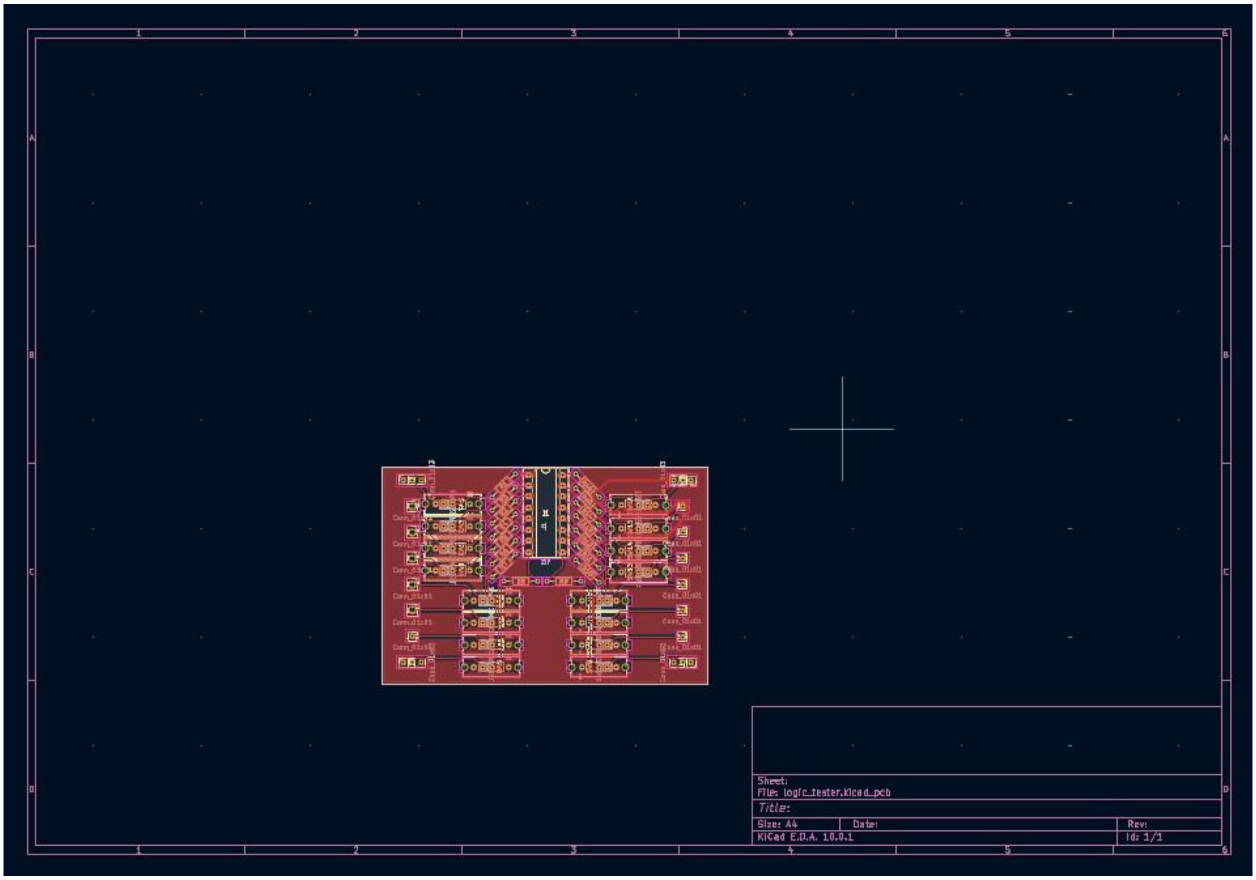


Layer 3:VCC plane

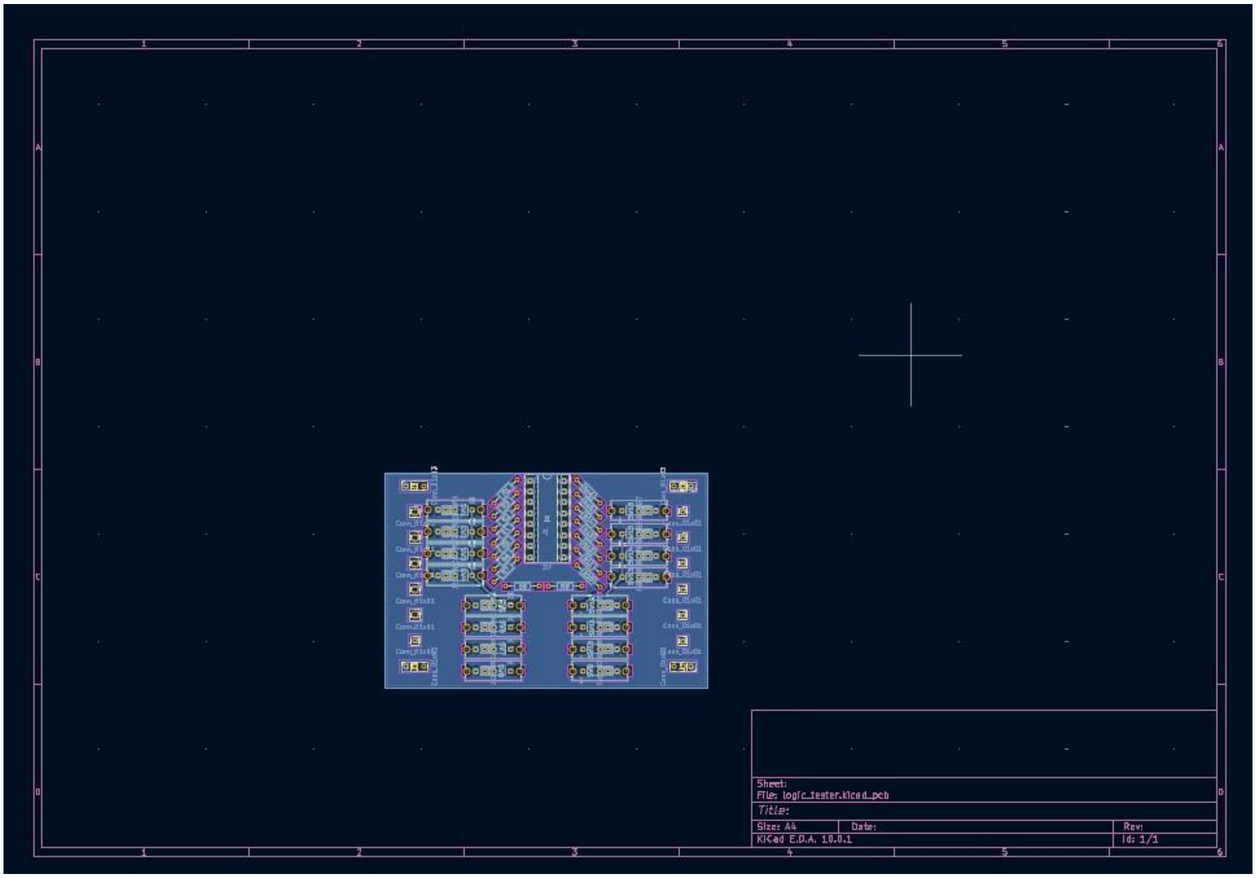


Layer 4:overflow layer

4.4 IC TESTER PCB

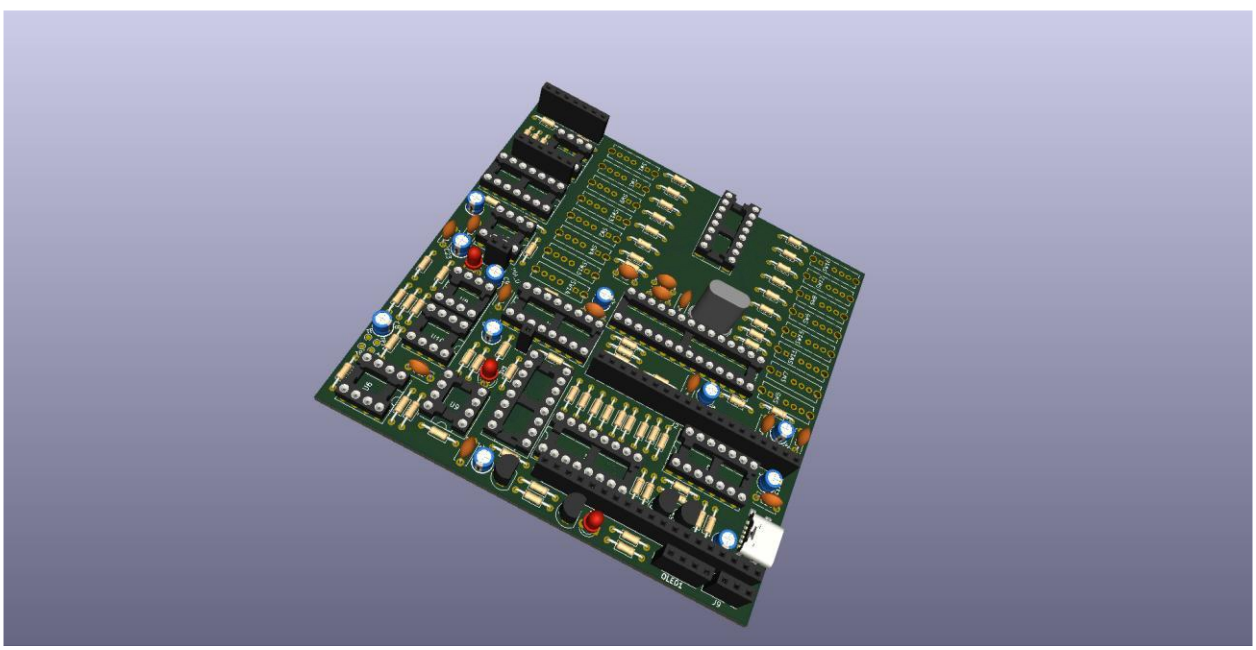


Layer 1:signal layer



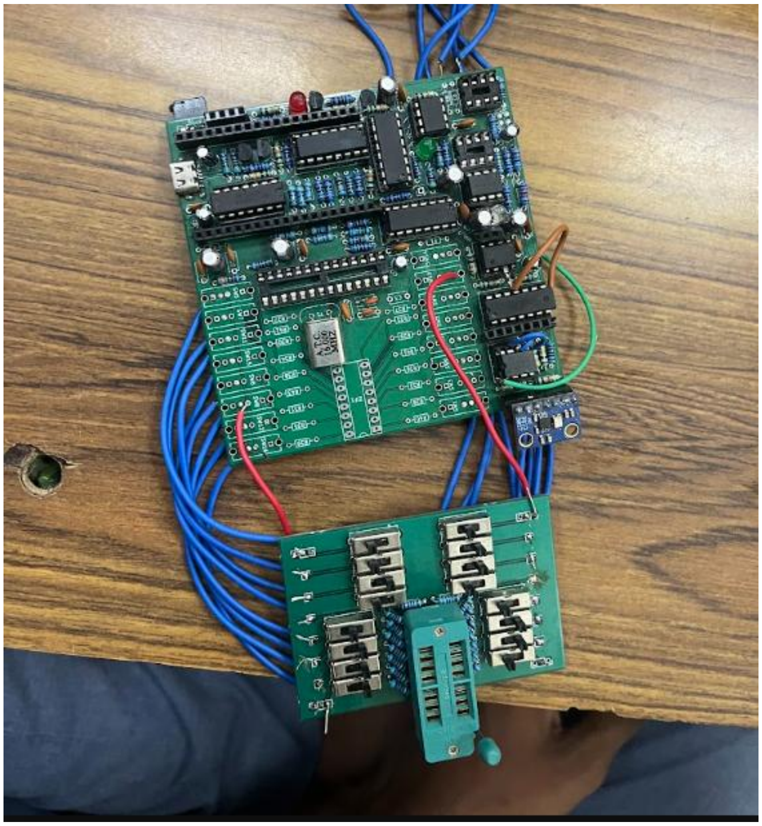
Layer 2:Ground Layer

5 Fabrication & Physical Assembly



3D viewer of the PCB

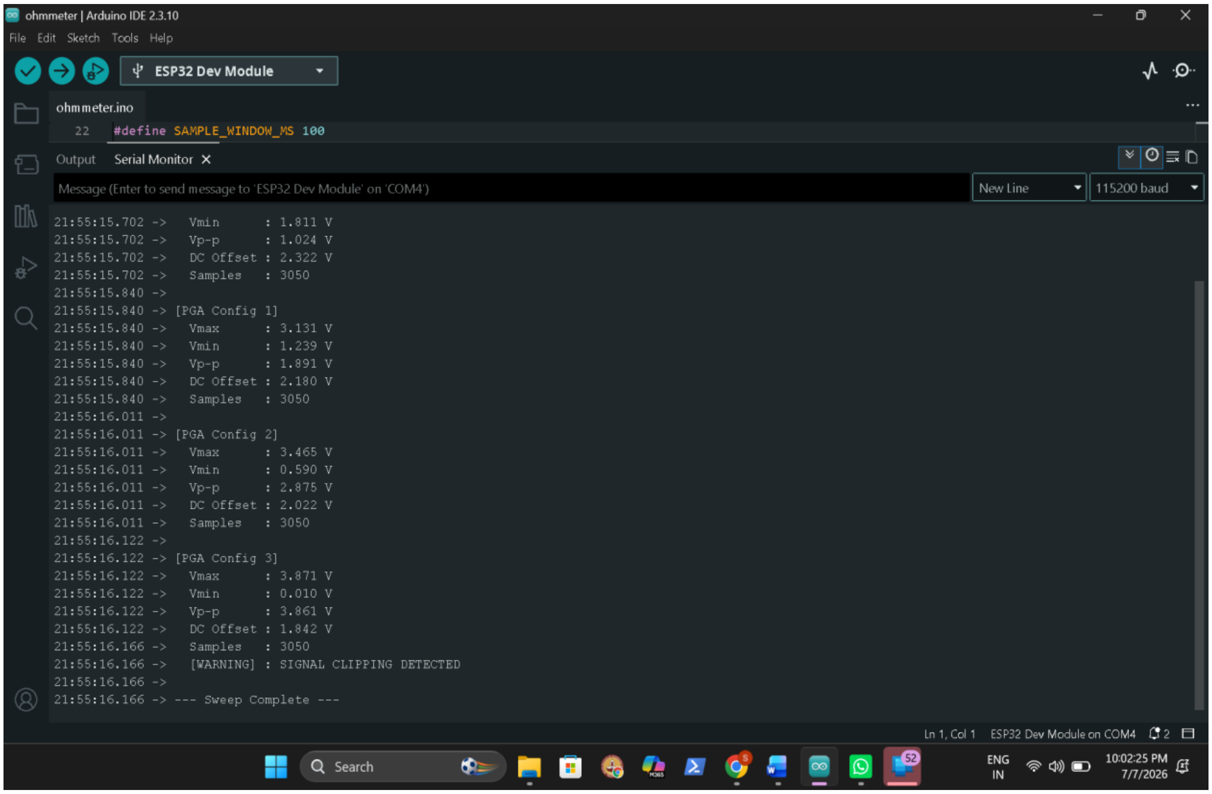
5.1 Hardware Debugging & Post-Assembly Challenges



The physical realization of Apex v1.0 on a 4-layer PCB successfully mitigated the noise and mechanical issues of the perfboard phase. However, as is standard with Version 1.0 hardware prototypes, post-assembly testing revealed several critical design flaws at both the silicon and routing levels that required manual intervention and firmware mitigation.

5.1.1 Analog Front-End: DC Bias Sag & Multiplexer R_{on}

During the initial PGA characterization, the system was programmed to sweep through the CD4052 multiplexer channels to verify the 2.5V nominal DC offset and peak-to-peak amplification. The serial telemetry revealed a severe, progressive sagging of the DC bias as the gain increased, ultimately resulting in signal clipping at the ground rail.



PGA Characterization Data (Sampled via MCP3208):

- **[PGA Config 0]:** $V_{p-p} = 1.024\text{V}$ | DC Offset = 2.322V
- **[PGA Config 1]:** $V_{p-p} = 1.891\text{V}$ | DC Offset = 2.180V
- **[PGA Config 2]:** $V_{p-p} = 2.875\text{V}$ | DC Offset = 2.022V
- **[PGA Config 3]:** $V_{p-p} = 3.861\text{V}$ | DC Offset = 1.842V *(Signal Clipping at* 0.010V*)*

5.1.2 Root Cause & Mitigation:

The investigation isolated the issue to the CD4052 analog multiplexer situated within the op-amp's feedback loop. The CD4052 is not an ideal switch; it possesses an internal on-resistance (R_{on}) of approximately 300Ω. At higher gain configurations, this parasitic resistance altered the impedance balance of the inverting amplifier topology, acting as an unintended voltage divider that pulled the DC bias downward. Because the signal clipped at Config 3, the firmware's auto-ranging algorithm was modified to detect saturation ($V_{min} < 0.05\text{V}$) and automatically step down to a lower gain stage, preventing invalid resistance calculations without requiring a silicon swap.

5.1.3 Microcontroller Integration: ESP32 Strapping Pin Conflicts

A major integration failure occurred during initial power-up, where the ESP32 master controller frequently failed to boot, entering a halted state instead of executing the firmware.

5.1.4 Root Cause & Mitigation:

The schematic design failed to account for the ESP32's specific "strapping pins" (e.g., GPIO 0, 2, 5, 12, 15). These pins sample their logic state at power-on to determine the IC's boot mode (e.g., executing from flash memory vs. entering serial download mode). During the PCB layout, several peripherals were inadvertently routed to these critical pins, pulling them to incorrect logic states during the boot sequence.

To resolve this without a costly secondary fabrication run, physical hardware modifications were executed. The copper traces leading to the offending strapping pins were manually severed using a precision blade.

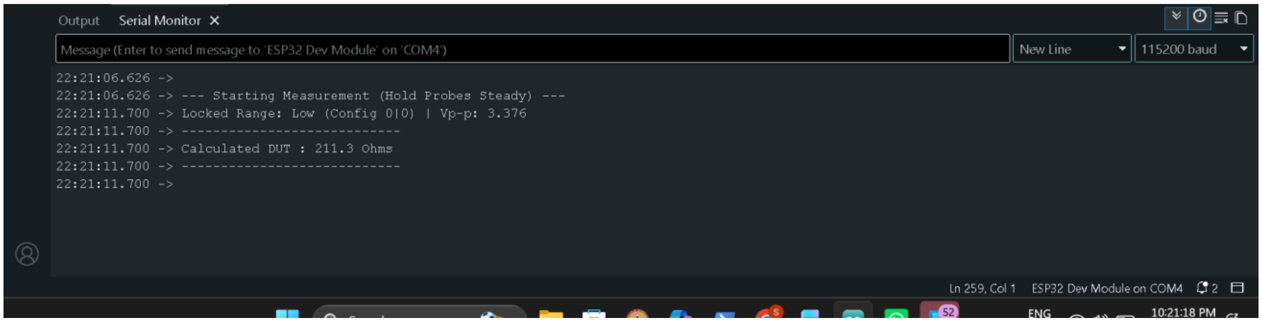
6 Routing Corrections and Manual PCB Patching

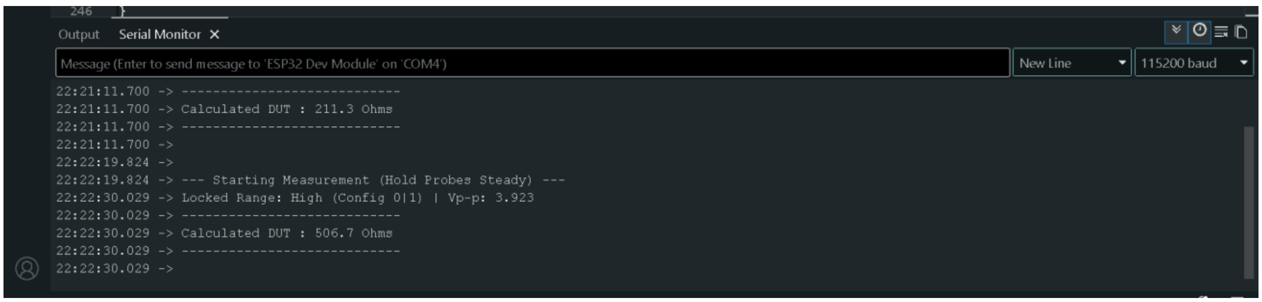
As this was a first-time high-density PCB layout, a few minor schematic-to-layout translation errors were discovered post-assembly (e.g., swapped logic lines and missed digital interconnects). Building on the strapping pin fixes, manual jumper wires (bodge wires) were soldered directly to the IC pads and vias to bridge the broken connections and reroute signals to safe, general-purpose ESP32 GPIOs. The firmware pin definitions were subsequently updated to reflect the patched hardware matrix. This manual intervention successfully salvaged the Version 1.0 prototype, yielding a fully operational diagnostic instrument.

7 System Validation & Results

The finalized Apex v1.0 hardware and firmware were subjected to rigorous functional validation to verify the auto-ranging algorithms and the digital diagnostic capabilities.

7.1 Auto-Ranging Ohmmeter Validation (0 to 3.3k Ohms)

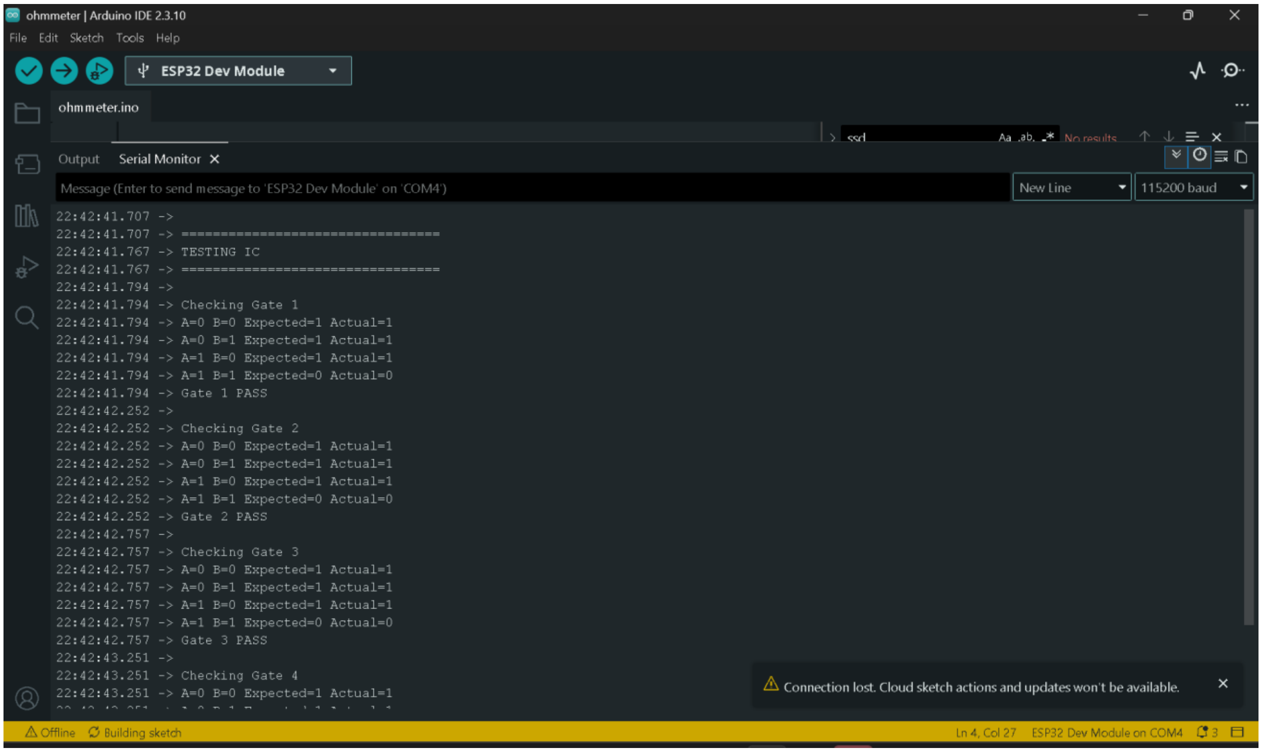


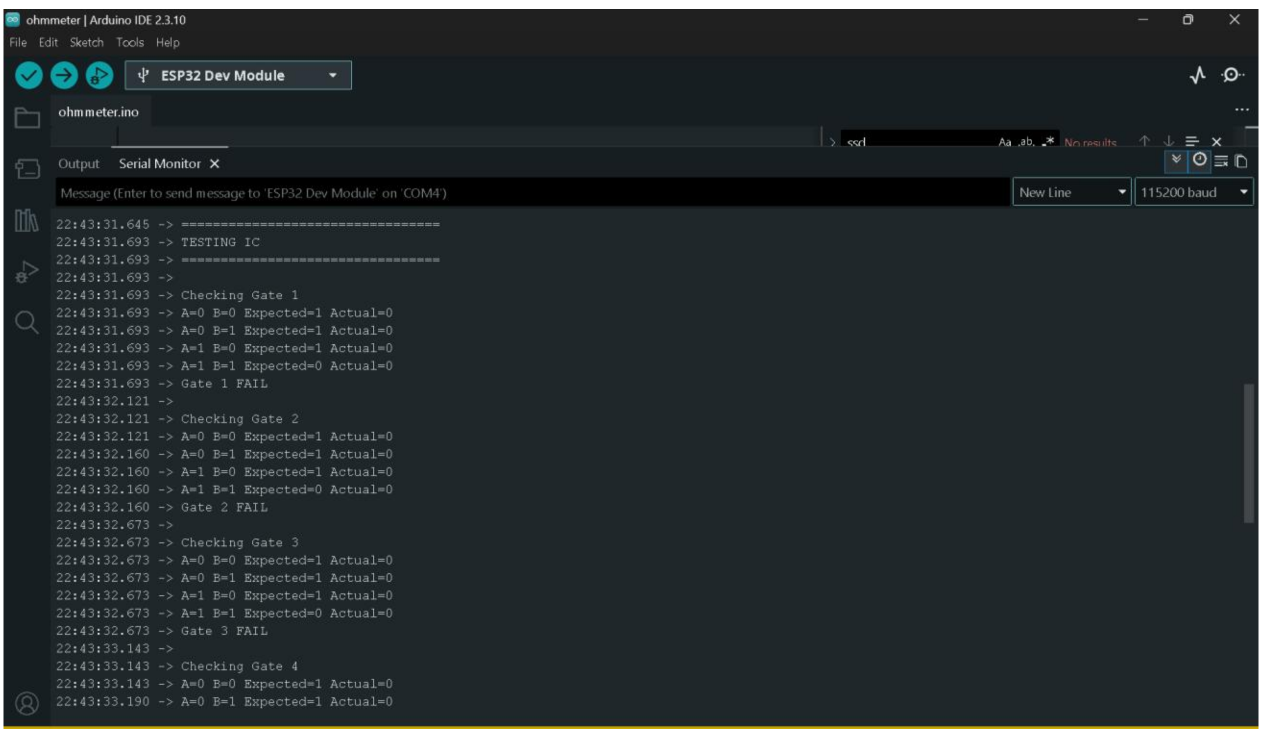


Due to the parasitic on-resistance (R_{on}) of the CD4052 multiplexer identified during the PGA characterization phase, operating the system at maximum gain introduced unacceptable DC offset sagging. To guarantee absolute measurement integrity and prevent ADC saturation, the operational envelope of the Version 1.0 prototype was deliberately constrained.

The firmware was locked to utilize only PGA Channel 0 as a stable, undistorted excitation baseline. By maintaining this static PGA configuration and dynamically auto-ranging the Transimpedance Amplifier (TIA) across Channel 0 and Channel 1, the system successfully established a highly accurate and continuous measurement band from **0 Ohms to 3.3k Ohms**. Within this restricted envelope, the firmware's mathematical interpolation engine executed flawlessly, successfully stepping between the low-gear and high-gear TIA feedback paths to calculate the impedance of standard test resistors with high repeatability.

7.2 Digital Logic IC Tester & Fault Injection Verification


ohmmeter | Arduino IDE 2.3.10
File Edit Sketch Tools Help
ESP32 Dev Module
ohmmeter.ino
Output Serial Monitor
Message (Enter to send message to 'ESP32 Dev Module' on 'COM4')
New Line
115200 baud
22:42:41.707 ->
22:42:41.707 -> ================================
22:42:41.767 -> TESTING IC
22:42:41.767 -> ================================
22:42:41.794 ->
22:42:41.794 -> Checking Gate 1
22:42:41.794 -> A=0 B=0 Expected=1 Actual=1
22:42:41.794 -> A=0 B=1 Expected=1 Actual=1
22:42:41.794 -> A=1 B=0 Expected=1 Actual=1
22:42:41.794 -> A=1 B=1 Expected=0 Actual=0
22:42:41.794 -> Gate 1 PASS
22:42:42.252 ->
22:42:42.252 -> Checking Gate 2
22:42:42.252 -> A=0 B=0 Expected=1 Actual=1
22:42:42.252 -> A=0 B=1 Expected=1 Actual=1
22:42:42.252 -> A=1 B=0 Expected=1 Actual=1
22:42:42.252 -> A=1 B=1 Expected=0 Actual=0
22:42:42.252 -> Gate 2 PASS
22:42:42.757 ->
22:42:42.757 -> Checking Gate 3
22:42:42.757 -> A=0 B=0 Expected=1 Actual=1
22:42:42.757 -> A=0 B=1 Expected=1 Actual=1
22:42:42.757 -> A=1 B=0 Expected=1 Actual=1
22:42:42.757 -> A=1 B=1 Expected=0 Actual=0
22:42:42.757 -> Gate 3 PASS
22:42:43.251 ->
22:42:43.251 -> Checking Gate 4
22:42:43.251 -> A=0 B=0 Expected=1 Actual=1
Connection lost. Cloud sketch actions and updates won't be available.
Offline
Building sketch
Ln 4, Col 27 ESP32 Dev Module on COM4


ohmmeter | Arduino IDE 2.3.10
File Edit Sketch Tools Help
ESP32 Dev Module
ohmmeter.ino
Output Serial Monitor
Message (Enter to send message to 'ESP32 Dev Module' on 'COM4')
New Line
115200 baud
22:43:31.645 -> ================================
22:43:31.693 -> TESTING IC
22:43:31.693 -> ================================
22:43:31.693 ->
22:43:31.693 -> Checking Gate 1
22:43:31.693 -> A=0 B=0 Expected=1 Actual=0
22:43:31.693 -> A=0 B=1 Expected=1 Actual=0
22:43:31.693 -> A=1 B=0 Expected=1 Actual=0
22:43:31.693 -> A=1 B=1 Expected=0 Actual=0
22:43:31.693 -> Gate 1 FAIL
22:43:32.121 ->
22:43:32.121 -> Checking Gate 2
22:43:32.121 -> A=0 B=0 Expected=1 Actual=0
22:43:32.160 -> A=0 B=1 Expected=1 Actual=0
22:43:32.160 -> A=1 B=0 Expected=1 Actual=0
22:43:32.160 -> A=1 B=1 Expected=0 Actual=0
22:43:32.160 -> Gate 2 FAIL
22:43:32.673 ->
22:43:32.673 -> Checking Gate 3
22:43:32.673 -> A=0 B=0 Expected=1 Actual=0
22:43:32.673 -> A=0 B=1 Expected=1 Actual=0
22:43:32.673 -> A=1 B=0 Expected=1 Actual=0
22:43:32.673 -> A=1 B=1 Expected=0 Actual=0
22:43:32.673 -> Gate 3 FAIL
22:43:33.143 ->
22:43:33.143 -> Checking Gate 4
22:43:33.143 -> A=0 B=0 Expected=1 Actual=0
22:43:33.190 -> A=0 B=1 Expected=1 Actual=0

To validate the ATmega328P co-processor and the 16-pin ZIF routing matrix, a targeted fault injection methodology was employed. The test parameters utilized standard 74-series logic chips, specifically quad 2-input NAND gates.

The validation was conducted in two phases:

1. **Baseline Verification:** A known-good NAND gate was seated in the ZIF socket, and the mechanical 1P3T switches were configured to provide VCC and GND. The ATmega328P successfully executed the bit-banged test vectors, verified the truth tables for all internal gates, and transmitted a "PASS" state to the ESP32 master.
2. **Destructive Fault Testing:** To verify the system's fault-detection capabilities, several NAND gate ICs were deliberately destroyed via overvoltage/overcurrent application, simulating severe field failures. When these faulted ICs were tested, the ATmega328P immediately identified the breakdown in the internal logic gates. The system successfully halted the test sequence and flagged the exact logic failures, proving the deterministic reliability of the IC tester and the mechanical switch routing matrix.

8 Conclusion & Key Learnings

The design and implementation of Apex v1.0 successfully demonstrated the viability of a low-cost, high-performance diagnostic instrument built on a custom dual-microcontroller architecture. By integrating an ESP32 master controller with an ATmega328P co-processor, the system effectively handled both high-speed SPI data acquisition and deterministic digital logic fault testing simultaneously. Moving from volatile breadboard prototypes to a dedicated 4-layer PCB was critical in minimizing electromagnetic interference and securing the signal integrity required for the 2.2V–2.8V analog front-end.

Key Engineering Learnings:

- **Silicon Realities vs. Theoretical Design:** The project highlighted the critical difference between ideal simulation and physical silicon. Encountering and mitigating the parasitic 300Ω on-resistance (R_{on}) of the CD4052 multiplexer underscored the importance of dynamic firmware algorithms to compensate for hardware limitations.
- **Boot-Strapping and Hardware Patching:** Managing the ESP32's strapping pin conflicts post-assembly provided practical experience in board-level debugging, trace cutting, and executing manual hardware patches (bodge wires) to salvage a Version 1.0 prototype.
- **Algorithmic Auto-Ranging:** Developing the firmware interpolation engine proved that complex, multi-gear analog scaling can be handled efficiently at the edge without requiring external processing power.

9.Future Scope & Version 2.0 Roadmap

While Apex v1.0 successfully established a highly accurate DC auto-ranging ohmmeter and a manually routed digital logic tester, the current dual-MCU architecture faces processing and bandwidth limitations. The development roadmap for Project Apex Version 2.0 focuses on a total architectural overhaul to transition the platform into a fully automated, mixed-signal diagnostic suite.

- **STM32 and AVR Architectural Upgrade:** Version 2.0 will replace the ESP32 master controller with a high-performance STM32 microcontroller. Utilizing the STM32's advanced DMA (Direct Memory Access) and high-speed hardware timers will eliminate the bottlenecks of SPI polling, allowing for significantly higher sampling rates. Dedicated AVR series MCUs will be retained as modular co-processors to handle isolated digital tasks.
- **Frequency Response Analysis (Bode Plotter):** The analog front-end will be upgraded to support full AC impedance and frequency response analysis. By sweeping the AD9833 signal generator across a wide frequency spectrum and calculating both gain attenuation and phase shift simultaneously, Version 2.0 will feature an integrated Bode plotter for evaluating complex reactive circuits and filter topologies.
- **Integrated Logic Analyzer:** Taking advantage of the increased STM32 clock speed and memory bandwidth, a multi-channel logic analyzer will be integrated to capture, decode, and visualize high-speed digital buses (I2C, SPI, UART) in real-time.
- **"Hands-Off" Autonomous IC Tester:** The mechanical 1P3T switch matrix used in Version 1.0 will be entirely replaced by a solid-state crossbar switching matrix. This fully automated ("hands-off") configuration will allow the master controller to dynamically route VCC, GND, and logic stimulus to any pin on the ZIF socket in milliseconds, enabling rapid, sequential batch testing of unknown ICs without physical user intervention.